

Ali Brooks

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Education

Rensselaer Polytechnic Institute (RPI)

Aug 2022 - May 2026

Bachelors of Science in Computer Science

Minors: Psychology

Relevant Courses: Computer Graphics, Graph Theory, Data Structures, Algorithms, Linear Algebra, Operating Systems

Experience

Blizzard Entertainment

May 2024 - August 2024

Gameplay Engineering Intern

Irvine, California

- Implemented new gameplay features with the gameplay engineering team for Diablo IV as well as immediate hotfixes and patches in shippable C++ code.
- Communicated with designers within every discipline to create tools and new accelerations that sped up their workflow for future content significantly.

Firaxis Games

May 2023 - August 2023

UI/UX Engineering Intern

Sparks, Maryland

- Built new UI/UX tools and features for Civilization VII in TypeScript, SCSS, HTML, and C++
- Navigated working with both Design and Art simultaneously to create long-lasting and modder friendly code.

Research

High Performance Combinatorics and Graph Analytics Lab

Aug 2025 - Present

Undergraduate Researcher

Troy, NY

- Designed and Programmed several high performance programs in C++/CUDA to run on the GPU
- Accelerated Graph I/O with a novel new GPU forward algorithm that significantly increased processing speeds on sparse graphs

BramHUB

Aug 2025 - Present

Lead Programmer

Troy, NY

<https://github.com/Bram-Hub/Sunflower>

- Worked directly with Professor Bram Van Heuveln as one of the lead programmers on a new website utilized by RPI
- Simplifies proofs related to computational recursion for use by students for homework submission
- Utilizes fast and efficient Typescript and React code

Perception and Action Lab (PANDA)

Jan 2023 - Dec 2023

Undergraduate Researcher

Troy, NY

- Collaborated with the Cognitive Science Department as the primary programmer on a Unity based simulation using C# under the guidance of Professor Brett Fajen
- Coordinated new virtual environments for eye-tracking experiments with experienced drone racing pilots as well as PANDA researchers.

Projects

Raytracing

<https://github.com/abrookst/Raytracing>

- Engineered a path tracing renderer from the ground up to simulate global illumination and physically-based rendering.
- Implemented diffuse material models to accurately simulate light interaction with surfaces.
- Integrated a Bounding Volume Hierarchy (BVH) to accelerate ray-object intersection tests, drastically improving performance.

Kinesis Engine

<https://github.com/abrookst/kinesis-engine>

- Developed a real-time ray tracing engine utilizing Vulkan for low-level GPU control and high performance.
- Architected a spectral rendering pipeline to achieve physically accurate color representation beyond standard RGB models.

Raycasting

<https://github.com/abrookst/raycasting>

- Designed and built a feature-complete 3D software renderer, processing all graphics calculations on the CPU.
- Utilized the SDL library for cross-platform window management and hardware surface interaction.

Skills

Software

Jira, Agile, Git, Perforce, Unity, Miro

Languages

C++, C, C#, Typescript, Python, Java, Rust

Soft

Collaboration, Problem Solving, Leadership, Cross Department Communication